



State of the Services FY 2024 Q3 Report

April 2024 - June 2024

Engagement Events Highlights

Industry Proving Grounds Task Group Meeting

Asheville, NC | June 11-13

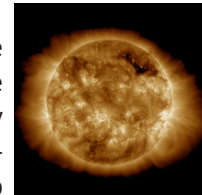
- The Inflation Reduction Act Industry Proving Grounds (IPG) Task Group met for a kick-off meeting where teams were familiarized with program goals, objectives, roles, and action items. These efforts in engagement, product development, and information delivery will promote resilience to climate risk and impact several sectors of the U.S. economy including Retail, Reinsurance, and Architecture & Engineering.
- Meetings with industry representatives and co-development of solutions continue, and has resulted in an upcoming 2024 NOAA IPG Hurricane Webinar.



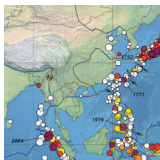
2024 Space Weather Workshop

Boulder, CO | April 15-19

- Twenty-three NCEI federal and affiliate staff participated in the 2024 Space Weather Workshop, co-organized by NOAA's Space Weather Prediction Center and several partners. The workshop is an annual conference that brings industry, academia, and government agencies together to engage in a lively dialog about space weather impacts in several areas, including communications, navigation, spacecraft operations, aviation, and electric power.
- NCEI staff presented five talks, five posters, and chaired one session.



Product and Services Highlights



Earthquake and Volcanic Eruption Impact Data Visualized

NCEI scientists collaborated with the International Tsunami Information Center to update the Significant Earthquakes and Significant Volcanic Eruptions posters. The Earthquake poster summarizes global earthquake data from significant events, including the fact that, since 1900, there have been 14 earthquakes with magnitude 8.6 or greater. The Volcanic Eruptions poster summarizes global volcano data from significant events, including the fact that 19 volcanic eruptions have generated deadly tsunamis.



NCEI Provides Critical Geomagnetic Latitude Information

NCEI provided the Federal Emergency Management Association (FEMA) with crucial geomagnetic latitude information during a significant geomagnetic storm occurring in May. FEMA used the data to assess the space weather hazard associated with the Severe/Extreme storm. NCEI furnished the necessary geomagnetic latitude information in the form of geospatial data based on the World Magnetic Model (WMM2020), which NCEI distributes on behalf of the National Geospatial-Intelligence Agency. The WMM2020, developed jointly by NCEI and the British Geological Survey, is a global spherical harmonic representation of the Earth's main magnetic field and its secular variation and is updated every five years to reflect Earth's magnetic field changes. It is critical for navigation and heading reference systems reliant on the geomagnetic field and serves as the standard for the U.S. Department of Defense, the United Kingdom Ministry of Defence, North Atlantic Treaty Organization, and the International Hydrographic Organization.



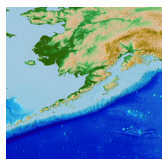
Wildfire Smoke Mortality Estimation Tool

The Northeast Regional Climate Center, along with a researcher from the Cornell Public Health Program, have developed a website that estimates mortality from wildfire smoke, by county, for the U.S. The site utilizes the U.S. Environmental Protection Agency's AirNow air quality sensor data and NOAA's Hazard Mapping System smoke plume polygons to identify those counties affected by wildfire smoke. It then uses a U.S. Centers for Disease Control and Prevention model to estimate the number of deaths that can be attributed to the adverse air quality resulting from the smoke plume.

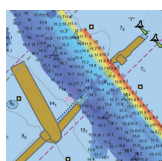
Stakeholder Highlights



NCEI participated in the American Association of State Climatologists (AASC) Annual Meeting in Fort Collins, CO and presented to this key partner community on the subjects of NCEI's strategic initiatives and product updates of interest, as well as understanding user needs related to projections of future extreme precipitation. One of the initiatives includes an upcoming AASC pilot activity to provide support to states as they support the goals of NCEI and NOAA for improving information delivery and societal outcomes.



Staff from Alaska Senator Lisa Murkowski's office met with NCEI and colleagues from the National Ocean Service's Office of Coast Survey and the Navy. The meeting request was triggered by the Senator's participation in the Navy's Ice Exercise - a biennial exercise conducted in the Beaufort Sea - where she was surprised at the lack of established mapping of the Arctic floor off the coast of Alaska (63% of U.S. waters off Alaska are unmapped). Senator Murkowski's staff were interested in learning more about the current U.S. efforts to improve this number, how these data are collected, made discoverable and accessible, and the current coordinated efforts between federal agencies.



NCEI posted hydrographic survey data for the limited-depth, limited-width federal channel reopening at the Francis Scott Key Bridge site in Baltimore, MD. The channel was temporarily closed after the bridge collapsed on March 26, 2024, when it was struck by a container ship. After the survey, review, and analysis by the Office of Coast Survey was completed, NCEI published the derived bathymetric surfaces and reports for public access.



NCEI was invited to participate in the Caribbean Climate Change Adaptation Summit in June, discussing NOAA's Equitable Climate Service Delivery Action Plan, with a focus on the steps NOAA is taking to advance resilience and adaptation through Caribbean-focused workforce development activities and improved data and information product delivery. The meeting was hosted by the Puerto Rico Climate Office and jointly sponsored by the University of Puerto Rico Mayaguez and the AASC with funding provided by NCEI.

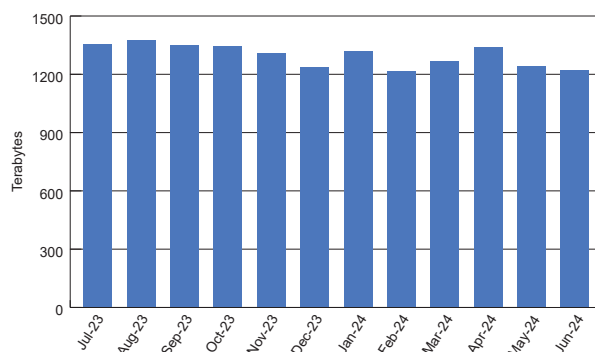
Upcoming Engagements

NOAA IPG Hurricane Webinar	Jul. 29
Extreme Heat Tabletop Exercise	Aug. 14-15
NOAA NextGen Satellite Conference	Sep. 10-12

Customer Reach

NCEI Makes Data Accessible

- 0.55 billion web hits and 3.8k TB of data delivered online*

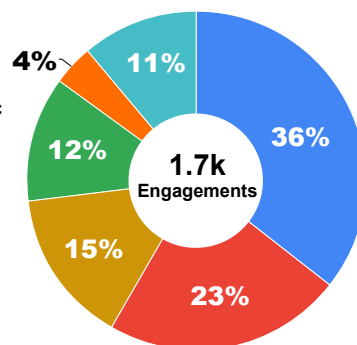


*Analytics include model & satellite data; data statistics may not be comprehensive across NCEI (e.g. CLASS)

U.S. Sectors Served

- Top Sectors

- Professional, Scientific, and Technical Services
- Other Services (except Public Administration)
- Educational Services
- Public Administration
- Finance and Insurance
- Others (Combined)



Unique Customer Requests

Alleviating Space Weather Anxiety

An autistic boy and his mother contacted NCEI regarding the boy's anxiety concerning geomagnetic storms and power cuts. News media coverage had led the boy to believe that the active sun would cause global power outage catastrophes. NCEI answered the boy's questions about forecasts, lead time, and how NOAA communicates with utility companies through the Space Weather Prediction Center. NCEI reassured the boy that there was no imminent threat from the sun.

Design Requirements for Water Tanks

An engineering firm contacted NCEI when determining water tank heating requirements for areas of the U.S. after seeing an increased demand for water tanks used for fire protection. NCEI provided access to the nClimGrid daily database in order to produce a customized climatology for the annual lowest minimum temperatures in the U.S. for various base periods. These data are critical to the design of these water tanks.

Fans



128.6k



65.4k



14.8k

